

Instructor Solution Notes

Companion to the first-semester Jupyter textbook.

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Circulation. These notes are for instructors. Do not place this file in the public student repository. Model answers are sufficient to mark; they are not the only acceptable wording. Award credit for the book's distinctions (split, stage, state versus content, credit to S6) even when the prose differs.

Marking stance. A correct slogan with a leaked split is not a pass. A cautious, incomplete number with a named split is closer to the book than a confident leaked one.

Chapter 0 — Refreshers

A1

$x = (1, 0, -1)$, $y = (2, 2, 0)$. Inner product = 2. $\|x\| = \sqrt{2}$. Cosine = $2 / (\sqrt{2} \cdot \sqrt{8}) = 2 / 4 = 0.5$. Accept equivalent calculation.

A2

Expect a 3×3 covariance with large equal variance on channels 1–2, large positive covariance between them, and near-zero cross terms with channel 3. Observation should match up to sampling noise.

A3

Set posterior = 1/2 at $x = 1.2$. Then $\text{like}_1 \cdot \text{prior}_1 = \text{like}_0 \cdot \text{prior}_0$. With $N(1.2; 0, 1) \approx 0.194$ and $N(1.2; 2, 1) \approx 0.290$, $\text{prior}_1 / \text{prior}_0 = 0.194 / 0.290 \approx 0.67$, so $\text{prior}_1 \approx 0.40$. Interpretation: the observation slightly favours class 1; a prior near 0.4 balances it. Accept 0.38–0.42.

A4

Overlapping windows share samples and slow state. A random fold puts near-duplicates on both sides. The score measures memorisation of a session, not generalisation.

A5

Ask: (i) What was the split — shuffled windows or later block? (ii) What is chance, and was an artefact/EMG decoder run? Calibration time and tomorrow's session are also fair.

B1

The ERP is small relative to ongoing EEG; averaging reduces uncorrelated noise. Cost for a live BCI: time. Many repetitions delay the character (Ch. 4–5).

B2

Correction: the implant records spikes and/or fields from a small volume, not a proposition. Decoding finds a statistical mapping from those measurements to a task-defined label.

B3

Neural: short-term synaptic change, fatigue, drying gel, unit drop-out. Strategy: the user found a new imagery or artefact policy after lunch.

B4

(i) single-cell event; (ii) population rhythm; (iii) population ERP visible after averaging.

B5

The claim assumes working-memory *content* is a usable scalp trace. Chapter 0–2 allow only mixed, blurred fields. At best a coarse state; content is not licensed.

Chapter 1 — What an interface is

1.1

Accept any ≤ 40 -word restatement that keeps measurement of CNS activity and artificial output. Lost if they drop research-tool use or the five purpose verbs — note the loss, do not fail solely for it.

1.2

(a) Non-invasive, brain→computer or passive, enhance/wellness, passive. Missing: whether an employer sees the score. (b) Invasive, brain→computer, replace, active. Missing: speech versus cursor. (c) Invasive or implanted, bidirectional, improve/restore, closed-loop stimulation.

1.3

Reactive: the generating event is the flash; the user attends. Active change: drop the matrix and ask for motor imagery or attempted speech on the same hardware (if the hardware can see it).

1.4

Different point on the resolution–risk surface; different information grain; no craniotomy does not make the signals equivalent. Consumer artefact and claim class also differ.

1.5

Must separate: possible EEG correlates of attempted or inner speech (weak); “cognitive computing” as branding; “decode inner speech” as a content claim without a usable trace. Six sentences that keep 8.1/8.3/8.4 pass.

1.6

Process: test the finger switch and gaze first; compare day-table communication; BCI only if those fail or fatigue them. Not automatic.

1.7

Cognitive computing: priors/policies. Neuromorphic: hardware style. Cognitive BCI: y is cognitive.

1.8

Any quoted headline plus a rewrite that names signal, task, and decoder. Fail if they only moralise.

Chapter 2 — First model of the brain**2.1**

Single-trial noise dwarfs the deflection; average reveals it. Cost: seconds of repetitions before a live decision.

2.2

Example: “The implant records electrical activity from one or a few nearby cells or a local field, and a decoder maps those measurements to a task label.”

2.3

(a) Yes, if it drives a switch she accepts. (b) No — group questionnaire correlate is not her options. (c) Yes, if the rate is one she will use.

2.4

Synaptic/metabolic drift or gel; strategy change. Both required.

2.5

Rate: higher firing for “reach right.” Population: direction lives in a pattern across cells. Mixed: one cell varies with both direction and grip.

2.6

Exploits implementational traces (ERP). Leaves algorithmic “how attention is computed” untouched. Accept Marr language used lightly.

2.7

Any four sources + weights with one weight ≈ 0 : that source can vanish at the electrode without vanishing in tissue.

Chapter 3 — Measuring and stimulating**3.1**

(a) low information, low medical risk; (b) modest information, low medical, high logistics; (c) mid information, surgical; (d) high information, high surgical and chronic risk. Two justifying sentences each.

3.2

Endovascular is a procedure. “Non-invasive” is false. Place it near ECoG on the surface, not next to a dry band.

3.3

Nyquist is 64 Hz. An 80 Hz feature cannot be represented. Anti-aliasing would have removed it.

3.4

Cochlear: computer→nerve, sensory prosthesis, not typically a BCI. Fixed DBS: stimulation without neural governance. Closed-loop BCI: measured activity governs the stimulus.

3.5

EEG families if vision allows; auditory ERP if not. Must test blink switch and gaze first.

3.6

Saturation, blanking, residual filter ringing. Strategies: blanking window, separate sites, template subtraction — outline only.

3.7

Must name explant funding, software successor, duration of support, what she keeps if she leaves. Six lines.

Chapter 4 — The loop

4.1

One sentence per stage. Language model at S6. Fail if they put it at S0.

4.2

Failure modes 3 and likely 4. Demand later-block closed-loop letters per minute with artefact audit.

4.3

Latency $\approx$ window + hop + 0.075 s. For 0.25/0.50/1.00 s windows with hop = 0.1·window: about 0.35, 0.625, 1.175 s plus the 75 ms extras already inside the notebook formula. A 2 Hz-wide μ feature is poorly estimated in 0.25 s (few cycles).

4.4

Definition in their words. Required: wheelchair wall constraint. Inflating: snap-to-icon counted as neural success.

4.5

Not finished: calibration consumes more than use.

4.6

Need EMG channels or forehead bipolar, nuisance decoder, cue-locked motion inspection. Halt if nuisance $\geq$ neural.

4.7

Employer or LMS is in S8. Other-directed loop.

4.8

Must add window, causal Φ , split, latency budget, user, intended use. Architecture name is not a design.

Chapter 5 — Paradigms

5.1

Eight answers: ERP to rares; reactive; EEG; time-locked shape; character; language prior possible; repetitions dominate latency; users who cannot see or attend.

5.2

Need covert-attention test, comparison to gaze-only, artefact audit, whether frequency tracks gaze.

5.3

If S0 has no usable ERD, a deeper net fits noise and EMG. Illiteracy is a generator problem.

5.4

Causal feature is dated at window end, so doubling the window adds about 0.75 s delay. That is loop latency.

5.5

Run S5 without LM; run LM on shuffled neural posteriors. Grammar that survives the second ablation is S6.

5.6

Imagery or auditory ERP remain. Still test a residual switch.

5.7

Muscle undo at S6/S8. Report neural accuracy excluding trials the muscle key decided.

5.8

Attempted speech has a motor-articulatory generator; inner speech is weaker and unsettled. No “reads thoughts.”

Chapter 6 — Features**6.1**

Referencing subtracts the blink from every channel, inventing a time-locked bump at all sites.

6.2

filtfilt is non-causal. Replace with causal band-pass; put group delay in the Chapter 4 budget.

6.3

90 percent overlap plus shuffle is leakage. Reject the number; demand block split.

6.4

First PC will likely align with blink or line — variance, not class. Not automatically neural.

6.5

CSP uses class covariances. Fit on all trials and the test class structure leaked. Nested: covariances from training blocks only.

6.6

Forehead EMG vs same labels, same split. Halt if EMG $\geq$ EEG.

6.7

(a) $z = \log \text{var}(w \cdot x_{\text{window}})$. (b) $z = [P(12 \text{ Hz}), P(24 \text{ Hz})]$. Accept equivalent.

6.8

Freeze montage, causal notch or not, window, no test-driven channel pick. Fan: consider 50 Hz handling before labels.

Chapter 7 — Learning**7.1**

Table of four drifts. Gap should widen with drift. Posters hide the causal column.

7.2

Chance is the floor of a coin. A baseline is a simpler f on the same split.

7.3

Against: parameters $\gg$ trials. Use rLDA / CSP+LDA.

7.4

Leave-one-user-out. Normalisation, CSP, early stopping, channel picks — all inside training users.

7.5

Remove LM; feed LM shuffled posteriors. Report both word-error numbers.

7.6

Undo = negative label. Failure: tired user stops undoing; errors reinforce.

7.7

Never-hit decoder: accuracy can look high because flashes are mostly non-targets; balanced accuracy near 0.5. Accept a small constructed table.

7.8

Rewrite must name split, chance, baseline. If the abstract cannot support that, say so.

Chapter 8 — Cognitive computing

8.1

8.1 language model on her prefix; 8.2 chip computing CSP in the cap; 8.3 a system whose y is her attention state. Six sentences.

8.2

Lower threshold when prior already favours the character. That is 8.1 at S6.

8.3

Title: “EEG classification of n-back load.” Result must drop shuffle or mark it invalid; call it state, not working-memory content.

8.4

Attempted speech has articulatory generators the implant can see. Inner-speech content is distributed, weakly labelled, easy to over-claim.

8.5

Protect: slow the matrix, wearer-directed. Rate: supervisor at S8, other-directed.

8.6

Hardware style at S4 does not evidence mixed selectivity in tissue. Different object (8.2 vs Ch. 2).

8.7

Two lines: “Claim class 8.1/8.2/8.3” and “state or content.”

8.8

No “reads thoughts / understands / brain-like.” Replace with estimate of a named state, on-device, wearer-directed.

Chapter 9 — Communication and motor

9.1

Compute both ITRs. Emergency yes/no prefers the fast binary even if ITR is lower. ITR ignores error cost.

9.2

12 cpm without setup/undos is not 9.1. Ask for household time.

9.3

Undo cell, blink, or dwell — S6/S8, not a new neural class.

9.4

Listener comprehension? Lag? Language of the prior? Credit split?

9.5

Table with S5 / speed cap / helper 10/40 / fail. Headline must not be 40/40.

9.6

Harm: output pulled toward the pretraining language. Mitigation: disable prior or train a prior in her language.

9.7

Plausible 20–40 min setup. Published cpm shrinks by the duty cycle. Accept arithmetic that includes setup.

9.8

Fixed target, helper DOF, no kettle dynamics. Tea-making needs more S6 and more S8.

Chapter 10 — Rehabilitation

10.1

(a) 10.2 restoration; (b) 10.3 improvement; (c) 10.1 / Ch. 9 replacement.

10.2

Same robot and therapist time; ■ shuffled or delayed so pairing breaks.

10.3

Consequence arrives after a long window; binding of intent to movement is weaker.

10.4

Therapist contact is an intervention. Confound.

10.5

Example: reduce robot assist 10 percent after three unassisted centimetres. Log assist fraction.

10.6

Sham trace; pre-registered attention test, not a vibe. Brochure cannot say treats ADHD.

10.7

Test residual mu/ERP on the available hemisphere. If absent, change family or do not enrol in imagery pairing.

10.8

Accuracy is process. Recovery is Definition 10.3.

Chapter 11 — Beyond the clinic**11.1–11.2**

Mark x by who sees ■. Rewrite “clinical-grade wellness” into either a medical claim with evidence or a wellness claim without disease language.

11.3

Local-only z and ■; no upload. List destinations: RAM, then deleted.

11.4

Hierarchy: signature ≠ free refusal. Six-line objection naming penalty.

11.5

Forehead EMG / brow effort. Nuisance decoder.

11.6

Wearer is the worker. Contract: purpose freeze, no sale, deletion, opt-out without feature loss if possible.

11.7

Amina+implant+XR is replace, wearer-directed. Exam band is other-directed evaluation.

11.8

Strip unlock/potential/read-minds. Leave named state, named sensor, named policy.

Chapter 12 — Evaluation, translation, ethics**12.1**

Twelve bullets covering the bundle for home P300. Missing owner-year-five may be N/A if non-implanted — they must say so.

12.2

Any abstract; list missing keys. Typical misses: split, artefact, credit, owner.

12.3

Intended use: attempted-speech communication in ALS. Not covered: inner-speech workplace monitoring.

12.4

Named hospital fund, successor firm or charity, explant offer, software escrow.

12.5

Weights of a spatial filter may still leak training structure. Strip identifiers; check licence; prefer filters fit on public de-identified sets.

12.6

Cannot use: photosensitive, low vision, no stable power/display, no caregiver for wet EEG, languages the UI lacks.

12.7

Rewrite wearer-directed with opt-out, or decline.

12.8

One-page protocol using bundle headings. Mark completeness, not ambition.

Appendix L — Laboratory write-ups

L1. Pass if they name a montage, justify or reject a notch, and admit the simulator is cleaner than a cafeteria.

L2. Pass if they report balanced accuracy and state that offline averaging is not a loop. Language-model sentence present.

L3. Headline number must be the causal split. Shuffled number mentioned only as a caution. EMG-audit gap named.

L4. Grade the sheet, not a promised accuracy. All sixteen lines attempted. A refused poster sentence is required.

Closing note to instructors

The book's standard is narrow on purpose. When two scripts collide, prefer the one that names the split, the stage, and the user. That is the masterpiece's method — not a longer adjective.